Apache Airflow

Exercise 2 — Student Marks Workflow DAG

student_marks_workflow_dag

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Program: Data Engineering | Batch: 2

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Topics: DAG Trigger • Graph View • Overview & Run History • Terminal Output

1. Exercise Overview

About This Exercise

This exercise demonstrates creating and running a student marks workflow DAG in Apache Airflow. The DAG reads student subject marks, calculates totals, and generates a result file — all orchestrated as a 5-step Python pipeline.

1.1 DAG Summary

Property	Value
DAG Name	student_marks_workflow_dag
Schedule	0 0 * * * (Daily at midnight)
Latest Run	2026-05-07 15:50:17 — Success
Next Run	2026-05-07 05:30:00
Max Active Runs	1 of 16
Owner	airflow | Version: v1
Failed Tasks (24h)	0
Operator Used	PythonOperator (all 5 tasks)

1.2 Task Pipeline

The DAG runs 5 tasks in a linear sequence using PythonOperator:

- create_marks_file — Creates student_marks.txt with subject-wise marks
- read_marks_file — Reads and parses marks from the file
- calculate_total — Sums all subject marks
- percentage_calculation — Calculates percentage from total
- generate_result — Writes final result summary to result.txt

```
create_marks_file >> read_marks_file >> calculate_total >> percentage_calculation
>> generate_result
```

2. Screenshot 1 — Trigger DAG Dialog

The trigger dialog is used to manually start a DAG run. It allows setting the Logical Date, data interval mode, and whether to auto-unpause the DAG on trigger.

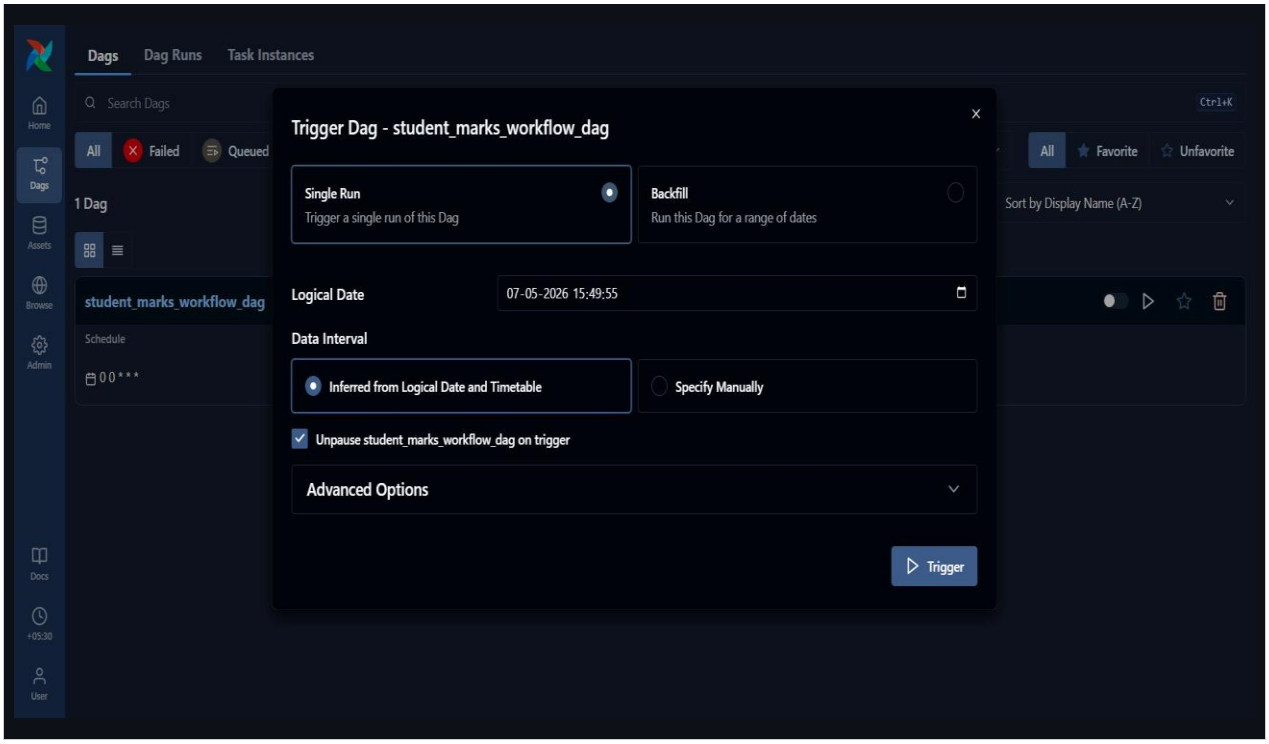

Figure 1: Trigger DAG dialog for student_marks_workflow_dag

Property	Value / Observation
DAG Name	student_marks_workflow_dag
Run Type	Single Run — Trigger a single run of this DAG
Logical Date	07-05-2026 15:49:55
Data Interval	Inferred from Logical Date and Timetable
Auto Unpause	Enabled — Unpause DAG on trigger
Trigger Mode	Manual trigger via Airflow UI

3. Screenshot 2 — DAG Overview & Run History

The Overview tab shows task completion status, run durations, and failure statistics for the last 24 hours. All 5 tasks show double green checkmarks indicating two consecutive successful runs.

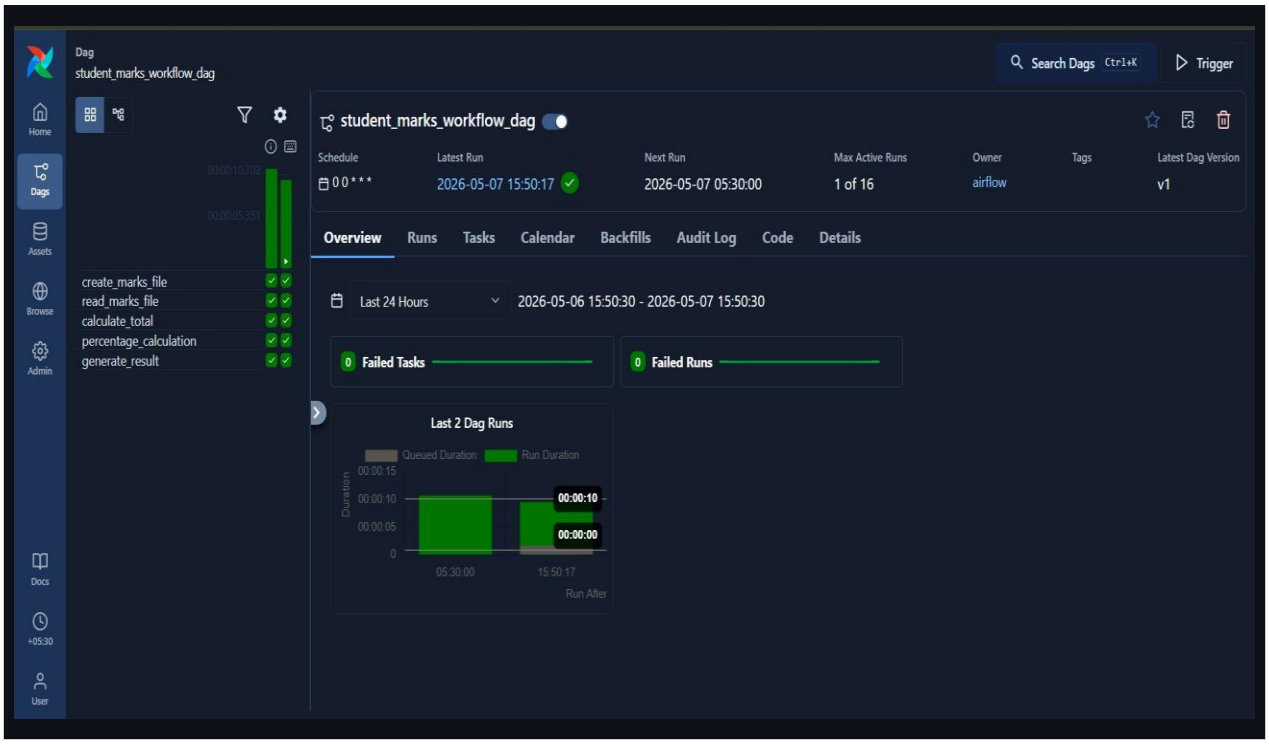

Figure 2: DAG Overview showing 5 tasks with double-success checkmarks

Property	Value / Observation
Tasks	create_marks_file, read_marks_file, calculate_total, percentage_calculation, generate_result
Task Status	All 5 — double green checkmark (2 consecutive successes)
Latest Run	2026-05-07 15:50:17 — Success
Run Duration	~10 seconds
Failed Tasks (24h)	0
Failed Runs (24h)	0
Runs Shown	Last 2 DAG runs: 05:30:00 and 15:50:17

4. Screenshot 3 — DAG Graph View

The Graph View visualises the DAG's task dependency structure. Each box is a task node and arrows represent the execution order — left to right in this linear pipeline.

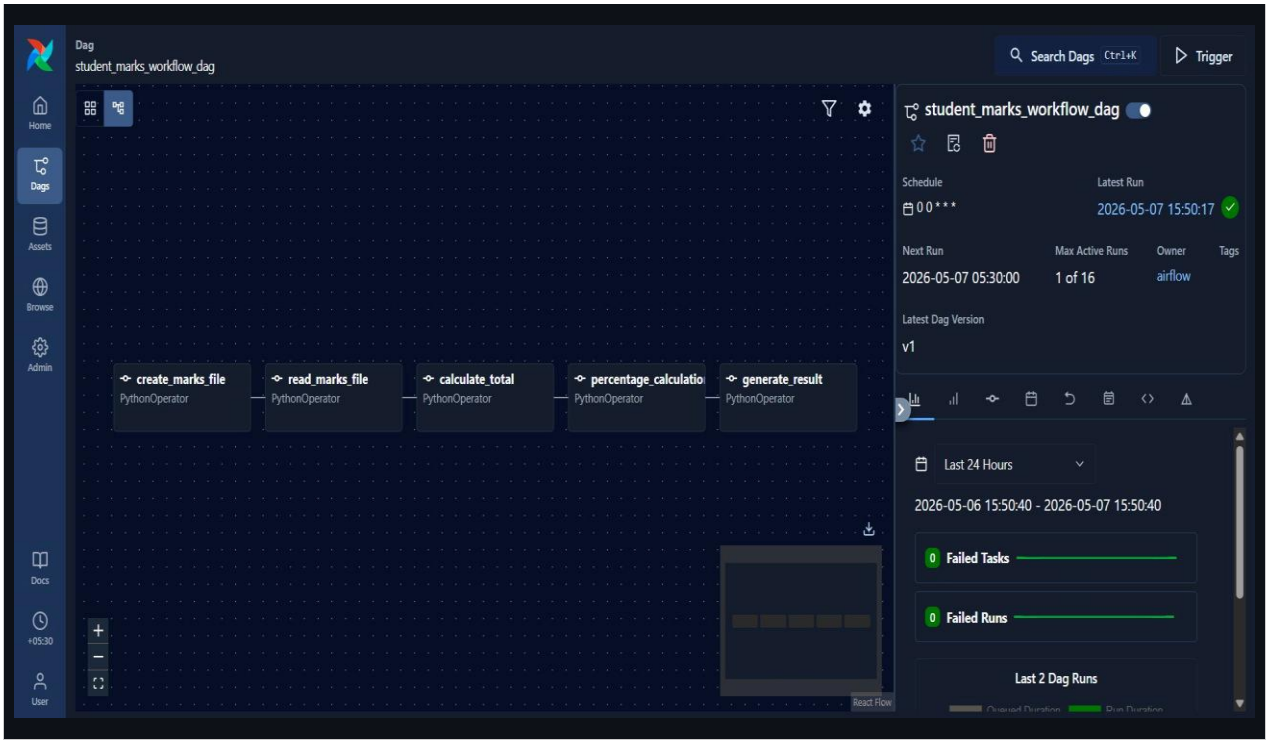

Figure 3: Graph View showing the 5-task linear pipeline

Property	Value / Observation
View	Graph View (React Flow)
Task 1	create_marks_file — PythonOperator
Task 2	read_marks_file — PythonOperator
Task 3	calculate_total — PythonOperator
Task 4	percentage_calculation — PythonOperator
Task 5	generate_result — PythonOperator
Dependency Type	Linear / sequential
Failed Tasks	0 — all green

5. Screenshot 4 — Terminal Output Verification

This terminal output from inside the Docker container confirms the DAG correctly created both the input marks file and the output result file in the /tmp directory.

```
root@ubuntu:~$ docker exec -it airflow bash
airflow@c71befb195ec:/opt/airflow$ cat /tmp/student_marks.txt
Math,80
Science,75
English,90
Python,95
airflow@c71befb195ec:/opt/airflow$ cat /tmp/result.txt
Student Result Summary
Total Marks = 340
Result = PASS
airflow@c71befb195ec:/opt/airflow$
```

Figure 4: Docker terminal showing student_marks.txt and result.txt

Property	Value / Observation
Command	docker exec -it airflow bash
Input File	/tmp/student_marks.txt
Subjects & Marks	Math: 80, Science: 75, English: 90, Python: 95
Output File	/tmp/result.txt
Total Marks	340 (80 + 75 + 90 + 95)
Result	PASS

5.1 Output Files Explained

student_marks.txt — Input File
Created by create_marks_file task. Contains: Math,80 / Science,75 / English,90 / Python,95 — comma-separated subject-mark pairs used by all downstream tasks.

result.txt — Output File

Generated by generate_result task. Contains: Student Result Summary / Total Marks = 340 / Result = PASS. Confirms the full pipeline ran end-to-end successfully.

5.2 Terminal Command

```
$ docker exec -it airflow bash
$ cat /tmp/student_marks.txt
Math,80  |  Science,75  |  English,90  |  Python,95
$ cat /tmp/result.txt
Student Result Summary  |  Total Marks = 340  |  Result = PASS
```

6. Summary

This exercise successfully demonstrated a complete end-to-end data workflow in Apache Airflow. The student_marks_workflow_dag read subject marks, computed totals, evaluated the result, and produced a verified output file — all with zero failures.

What Was Demonstrated	Outcome
Manual DAG Trigger	Triggered via UI dialog successfully
5-task linear pipeline	All tasks completed with PythonOperator
File I/O operations	marks file created, read, result written
Total marks calculation	Math+Science+English+Python = 340
Pass/Fail logic	Result = PASS confirmed in result.txt
Zero failures	0 failed tasks, 0 failed runs in 24h

— End of Exercise 2 —